Solutions to Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed.)

Part I: One-Dimensional Flows

Chapter 4: Flows on the Circle

Conventions. On circle portraits, angles are measured counterclockwise from the positive x-axis; a *filled* black dot is a stable fixed point, an *open* dot an unstable one, and red arrows give the direction of flow. For the singular vector fields of 4.3.4, 4.3.7 and 4.3.8, a *red diamond* marks a point where $|\dot\theta| \to \infty$: a filled diamond is an *attracting* singularity (trajectories on both sides run into it), an open diamond a *repelling* one. Problem statements are abbreviated.

4.1 Examples and Definitions

4.1.1 *For which real a does $\dot\theta = \sin(a\theta)$ define a vector field on the circle?*

Solution. A vector field on the circle must be single-valued: $f(\theta + 2\pi) = f(\theta)$ for all θ, i.e. $\sin(a\theta + 2\pi a) = \sin(a\theta)$ for all θ. Taking $\theta = 0$ gives $\sin 2\pi a = 0$, so $2a \in \mathbb{Z}$; taking $\theta = \pi/(2a)$ (for $a \neq 0$) gives $\cos 2\pi a = 1$, so $a \in \mathbb{Z}$. Conversely, every integer a works. Hence a **must be an integer** ($a = 0$ gives the trivial field $\dot\theta = 0$). ■

4.1.2–4.1.7 *Find and classify all fixed points and sketch the phase portrait on the circle.*

Solution. Stability on the circle is decided exactly as on the line: $f'(\theta^*) > 0$ unstable, $f'(\theta^*) < 0$ stable (Fig. 1).

4.1.2 $\dot\theta = 1 + 2\cos\theta$**.** $\cos\theta = -\frac{1}{2} \Rightarrow \theta^* = \pm\frac{2\pi}{3}$. $f' = -2\sin\theta$: $f'(\frac{2\pi}{3}) = -\sqrt{3} < 0$ (**stable**), $f'(-\frac{2\pi}{3}) = +\sqrt{3} > 0$ (**unstable**).

4.1.3 $\dot\theta = \sin 2\theta$**.** $\theta^* = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. $f' = 2\cos 2\theta$: $\theta^* = 0, \pi$ are **unstable** ($f' = 2$), $\theta^* = \frac{\pi}{2}, \frac{3\pi}{2}$ are **stable** ($f' = -2$).

4.1.4 $\dot\theta = \sin^3\theta$**.** $\theta^* = 0, \pi$. Here $f' = 3\sin^2\theta\cos\theta = 0$ at both, so use the sign of f: $\sin^3\theta$ has the sign of $\sin\theta$, positive on $(0, \pi)$ and negative on $(\pi, 2\pi)$. So $\theta^* = 0$ is **unstable** and $\theta^* = \pi$ is **stable**: qualitatively the same portrait as $\dot\theta = \sin\theta$ (with algebraic rather than exponential approach to π).

4.1.5 $\dot\theta = \sin\theta + \cos\theta = \sqrt{2}\sin(\theta + \frac{\pi}{4})$**.** $\theta^* = -\frac{\pi}{4}$ and $\frac{3\pi}{4}$. $f' = \sqrt{2}\cos(\theta + \frac{\pi}{4})$: $f'(-\frac{\pi}{4}) = \sqrt{2} > 0$ (**unstable**), $f'(\frac{3\pi}{4}) = -\sqrt{2} < 0$ (**stable**).

4.1.6 $\dot\theta = 3 + \cos 2\theta$**.** $3 + \cos 2\theta \geq 2 > 0$: **no fixed points**; the flow is a nonuniform counterclockwise rotation (fastest at $\theta = 0, \pi$, slowest at $\pm\frac{\pi}{2}$). Its period is $T = \int_0^{2\pi} \frac{\mathrm{d}\theta}{3+\cos 2\theta} = \frac{2\pi}{\sqrt{9-1}} = \frac{\pi}{\sqrt{2}}$.

4.1.7 $\dot\theta = \sin k\theta$ **(k a positive integer).** $k\theta^* = m\pi$, so $\theta^* = m\pi/k$, $m = 0, 1, \ldots, 2k-1$: there are $2k$ fixed points. $f' = k\cos k\theta$ gives $f'(m\pi/k) = k(-1)^m$: those with m even are **unstable**, those with m odd **stable**; stable and unstable points alternate around the circle (k of each).

■

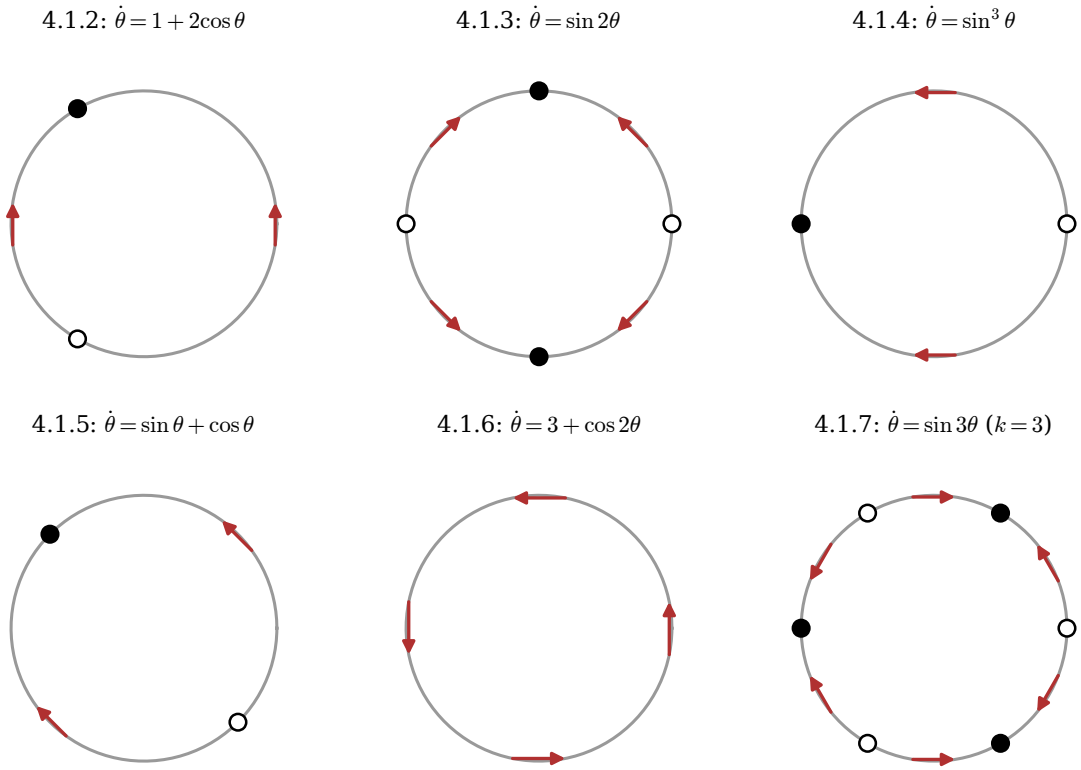


Figure 1: Phase portraits for 4.1.2–4.1.7 (4.1.7 drawn for $k = 3$).

4.1.8 *(Potentials on the circle) (a) Show that* $\dot\theta = \cos\theta$ *has a single-valued potential. (b) Show that* $\dot\theta = 1$ *does not. (c) What is the general rule?*

Solution. **(a)** $V(\theta) = -\sin\theta$ satisfies $\dot\theta = -\mathrm{d}V/\mathrm{d}\theta = \cos\theta$, and $V(\theta + 2\pi) = V(\theta)$, so V is a well-defined function on the circle.

(b) Any potential must be $V(\theta) = -\theta + \text{const}$, and then $V(\theta + 2\pi) = V(\theta) - 2\pi \neq V(\theta)$: the value of V after going once around the circle differs from its starting value, so V is not single-valued. (Equivalently: $\oint V'\,\mathrm{d}\theta$ must be 0, but $\oint(-1)\,\mathrm{d}\theta = -2\pi$.)

(c) $V(\theta) = -\int_0^\theta f(\phi)\,\mathrm{d}\phi$ is 2π-periodic iff $V(2\pi) = V(0)$, i.e.

$$\boxed{\int_0^{2\pi} f(\theta)\,\mathrm{d}\theta = 0}:$$

$\dot\theta = f(\theta)$ has a single-valued potential exactly when f has zero average around the circle. In particular, any f with a nonzero mean (such as a flow with no fixed points, which has constant sign) has none. ■

4.1.9 *Explain why the two proofs (2.6.2 and 2.7.7) that periodic solutions are impossible on the line do not carry over to the circle.*

Solution. *Proof of 2.6.2.* It used $\int_t^{t+T} f(x)\dot{x}\,\mathrm{d}t = \int_{x(t)}^{x(t+T)} f\,\mathrm{d}x = 0$ because $x(t+T) = x(t)$. On the circle a periodic solution can go once around: $\theta(t+T) = \theta(t) + 2\pi$, which is the *same point* on the circle but a *different real number*. The integral is then $\int_0^{2\pi} f(\theta)\,\mathrm{d}\theta$, which need not vanish (indeed it is positive for $\dot{\theta} = f > 0$), so there is no contradiction with $\int \dot{\theta}^2 \mathrm{d}t > 0$. The step that fails is the equality of the end-point values of the antiderivative.

Proof of 2.7.7. It used a potential V with $V(x(t+T)) = V(x(t))$, so that $\Delta V = 0$ over a period, versus $\mathrm{d}V/\mathrm{d}t = -\dot{x}^2 < 0$. On the circle V need not be single-valued (4.1.8): for $\dot{\theta} = 1$ there is no potential on the circle at all, and when one lifts to $V(\theta) = -\theta$ on the line, V decreases by 2π per revolution, exactly consistent with $\dot{V} < 0$. The step that fails is the existence of a single-valued potential, i.e. $\oint f\,\mathrm{d}\theta = 0$. (If f *does* have a single-valued potential on the circle, the argument goes through and there are no periodic solutions.) ■

4.2 Uniform Oscillator

4.2.1 *(Church bells) One bell rings every 3 s and the other every 4 s; they just rang together. When next?*

Solution. *Common sense.* The bells coincide again at the first common multiple of 3 and 4: $\mathrm{lcm}(3,4) = \boxed{12\text{ s}}$ (after 4 rings of the first and 3 of the second).

Method of Example 4.2.1. The phases are $\theta_1 = 2\pi t/3$ and $\theta_2 = 2\pi t/4$. The bells ring together when the phase difference returns to 0 modulo 2π:

$$\theta_1 - \theta_2 = 2\pi t\Big(\frac{1}{3} - \frac{1}{4}\Big) = \frac{2\pi t}{12} = 2\pi \quad\Longrightarrow\quad t = \frac{T_1 T_2}{T_2 - T_1} = \frac{3\cdot 4}{4-3} = 12\text{ s}.$$

At $t = 12$, $\theta_1 = 8\pi$ and $\theta_2 = 6\pi$, so both are at the ringing phase. ■

4.2.2 *(Beats) Graph $x(t) = \sin 8t + \sin 9t$ for $-20 < t < 20$. (a) Period of the amplitude modulation? (b) Solve analytically with a trigonometric identity.*

Solution. **(b)** Using $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$,

$$x(t) = 2\cos\Big(\frac{t}{2}\Big)\,\sin\Big(\frac{17t}{2}\Big).$$

This is a fast carrier $\sin(17t/2)$ (period $4\pi/17$) multiplied by a slow envelope $2\cos(t/2)$.

(a) The amplitude is $|2\cos(t/2)|$, which grows and decays periodically (Fig. 2); each maximum of $|2\cos(t/2)|$ is a "beat". Since $|\cos(t/2)|$ has period 2π, the beats occur every $\boxed{2\pi \approx 6.28}$ time units, i.e. at the beat frequency $9 - 8 = 1$ (radians per unit time; cycles per unit time $1/2\pi$). (The signal itself also repeats every 2π, because $\mathrm{lcm}(2\pi/8,\, 2\pi/9) = 2\pi$.) For tuning an instrument: the closer the two frequencies, the longer the time between beats. ■

4.2.3 *(Clock problem) At 12:00 the hour and minute hands coincide. When next?*

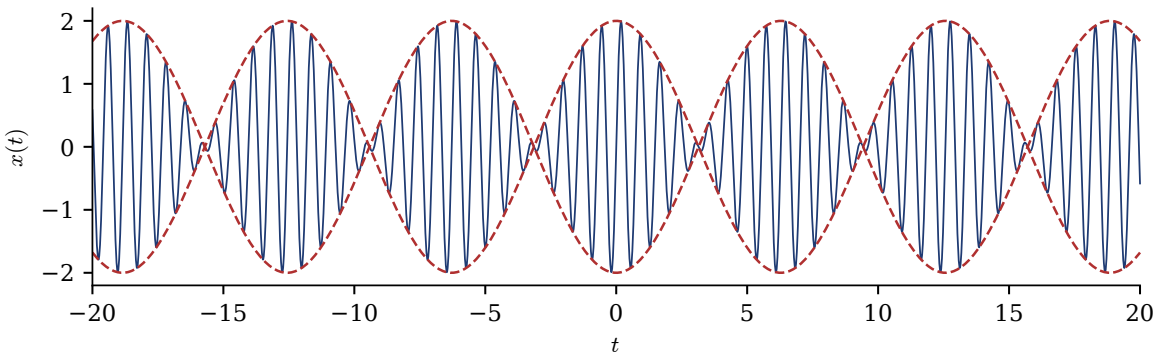


Figure 2: $x(t) = \sin 8t + \sin 9t$ (solid) and the envelope $\pm 2\cos(t/2)$ (dashed).

Solution. The minute hand has period 1 h and the hour hand 12 h, so their angular speeds are 2π and $2\pi/12$ rad/h. By the method of this section, the hands align when the relative phase advances by 2π:

$$\Big(2\pi - \frac{2\pi}{12}\Big)t = 2\pi \quad \Longrightarrow \quad t = \frac{12}{11}\text{ h} = 1\text{ h }5\tfrac{5}{11}\text{ min} \approx 1{:}05{:}27.3.$$

Alternative. In degrees, after t hours the minute hand is at $360t$ and the hour hand at $30t$; they coincide when $360t - 30t = 360k$, so $t = 12k/11$; $k = 1$ gives the next time. (This also shows the hands coincide 11 times in 12 hours, at $t = 12k/11$ h, $k = 0, \dots, 10$.) ■

4.3 Nonuniform Oscillator

4.3.1 *Evaluate* $T_{\text{bottleneck}} = \int_{-\infty}^{\infty} \frac{dx}{r+x^2}$ *with* $x = \sqrt{r}\tan\theta$.

Solution. $dx = \sqrt{r}\sec^2\theta\, d\theta$ and $r + x^2 = r(1 + \tan^2\theta) = r\sec^2\theta$, so the integrand becomes $\frac{\sqrt{r}\sec^2\theta}{r\sec^2\theta}d\theta = \frac{d\theta}{\sqrt{r}}$. As x runs from $-\infty$ to ∞, θ runs from $-\pi/2$ to $\pi/2$:

$$T_{\text{bottleneck}} = \frac{1}{\sqrt{r}}\int_{-\pi/2}^{\pi/2} d\theta = \frac{\pi}{\sqrt{r}}.$$

■

4.3.2 *Evaluate* $T = \int_{-\pi}^{\pi} \frac{d\theta}{\omega - a\sin\theta}$ *($\omega > a > 0$) by the substitution* $u = \tan\frac{\theta}{2}$.

Solution. **(a)** $\theta = 2\tan^{-1}u$, so $d\theta = \dfrac{2\,du}{1+u^2}$.

(b) In a right triangle with base 1 and height u, $\theta/2$ is the angle opposite the side u; the hypotenuse is $\sqrt{1+u^2}$, so $\sin\frac{\theta}{2} = \frac{u}{\sqrt{1+u^2}}$, $\cos\frac{\theta}{2} = \frac{1}{\sqrt{1+u^2}}$, and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \dfrac{2u}{1+u^2}$.

(c) As $\theta \to \pi^-$, $\theta/2 \to \pi/2^-$ and $u \to +\infty$; as $\theta \to -\pi^+$, $u \to -\infty$. So the limits become $-\infty$ to ∞.

(d) Substituting,

$$T = \int_{-\infty}^{\infty} \frac{2\,du/(1+u^2)}{\omega - \frac{2au}{1+u^2}} = \int_{-\infty}^{\infty} \frac{2\,du}{\omega(1+u^2) - 2au} = \int_{-\infty}^{\infty} \frac{2\,du}{\omega u^2 - 2au + \omega}.$$

(e) Complete the square: $\omega u^2 - 2au + \omega = \omega\big[(u - \frac{a}{\omega})^2 + 1 - \frac{a^2}{\omega^2}\big]$. With $x = u - a/\omega$ and $r = 1 - a^2/\omega^2 > 0$,

$$T = \frac{2}{\omega}\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2 + r} = \frac{2}{\omega}\cdot\frac{\pi}{\sqrt{r}} = \frac{2\pi}{\omega\sqrt{1 - a^2/\omega^2}} = \boxed{\frac{2\pi}{\sqrt{\omega^2 - a^2}}}$$

by 4.3.1. ■

4.3.3–4.3.8. For each system I find the fixed points and their stability as functions of μ, identify the bifurcations, and show circle portraits for representative μ together with the bifurcation diagram (θ^* versus μ; black = stable, grey = unstable, red = singular points). For the three quotient fields (4.3.4, 4.3.7, 4.3.8) the denominator can vanish, where $|\dot\theta| \to \infty$ and the equation does not define a flow; such points are marked as singularities, not fixed points. Near a pole $\dot\theta \approx c/(\theta_s - \theta)$, so a trajectory reaches it in finite time. A useful symmetry for 4.3.4 and 4.3.7: since $f_\mu(\theta + \pi) = f_{-\mu}(\theta)$, the portrait for $-\mu$ is the portrait for μ rotated by π.

4.3.3 $\dot\theta = \mu\sin\theta - \sin 2\theta$.

Solution. $f = \sin\theta\,(\mu - 2\cos\theta)$. Fixed points: $\theta^* = 0, \pi$ for all μ, and $\cos\theta^* = \mu/2$, i.e. $\theta^* = \pm\alpha$, $\alpha = \cos^{-1}(\mu/2)$, which exist only for $|\mu| < 2$. With $f'(\theta) = \mu\cos\theta - 2\cos 2\theta$:

- $f'(0) = \mu - 2$: $\theta^* = 0$ is stable for $\mu < 2$, unstable for $\mu > 2$.
- $f'(\pi) = -\mu - 2$: $\theta^* = \pi$ is stable for $\mu > -2$, unstable for $\mu < -2$.
- $f'(\pm\alpha) = \mu\cdot\frac{\mu}{2} - 2\big(\frac{\mu^2}{2} - 1\big) = 2 - \frac{\mu^2}{2} = \frac{4-\mu^2}{2} > 0$: the extra fixed points $\pm\alpha$ are always **unstable**.

So for $|\mu| < 2$ there are four fixed points ($0, \pi$ stable; $\pm\alpha$ unstable); for $\mu > 2$ only 0 (unstable) and π (stable); for $\mu < -2$ only 0 (stable) and π (unstable).

Bifurcations: at $\mu = 2$ the unstable points $\pm\alpha \to 0$ as $\mu \to 2^-$ and merge into $\theta = 0$, which loses stability: a **subcritical pitchfork** at $(\theta, \mu) = (0, 2)$. At $\mu = -2$, $\pm\alpha \to \pi$: another **subcritical pitchfork** at $(\pi, -2)$. (Both reflect the symmetry $f(-\theta) = -f(\theta)$.) ■

4.3.4 $\dot\theta = \dfrac{\sin\theta}{\mu + \cos\theta}$.

Solution. The zeros are $\theta = 0, \pi$ (where $\mu + \cos\theta = \mu \pm 1 \neq 0$ unless $\mu = \mp 1$). Differentiating, $f'(\theta) = \dfrac{1 + \mu\cos\theta}{(\mu + \cos\theta)^2}$, so

$$f'(0) = \frac{1}{1+\mu}, \qquad f'(\pi) = \frac{1}{1-\mu}.$$

- $\mu > 1$: $f'(0) > 0$, $f'(\pi) < 0$; the denominator is positive, the field is smooth: $\theta^* = 0$ **unstable**, $\theta^* = \pi$ **stable**.
- $\mu < -1$: smooth; $\theta^* = 0$ **stable**, $\theta^* = \pi$ **unstable** (rotation of the previous portrait by π).
- $|\mu| < 1$: $\mu + \cos\theta$ vanishes at $\theta_s = \pm\cos^{-1}(-\mu)$, so the field is **singular** there. Both $f'(0) = \frac{1}{1+\mu} > 0$ and $f'(\pi) = \frac{1}{1-\mu} > 0$: both 0 and π are **unstable**. Near $\theta_s \in (0, \pi)$ (where $\sin\theta > 0$) the denominator changes from $+$ to $-$, so $\dot\theta \to +\infty$ from the left and $-\infty$ from the right: each singularity is **attracting**. Hence the flow is: leave 0 and π and run into the nearest singular points.

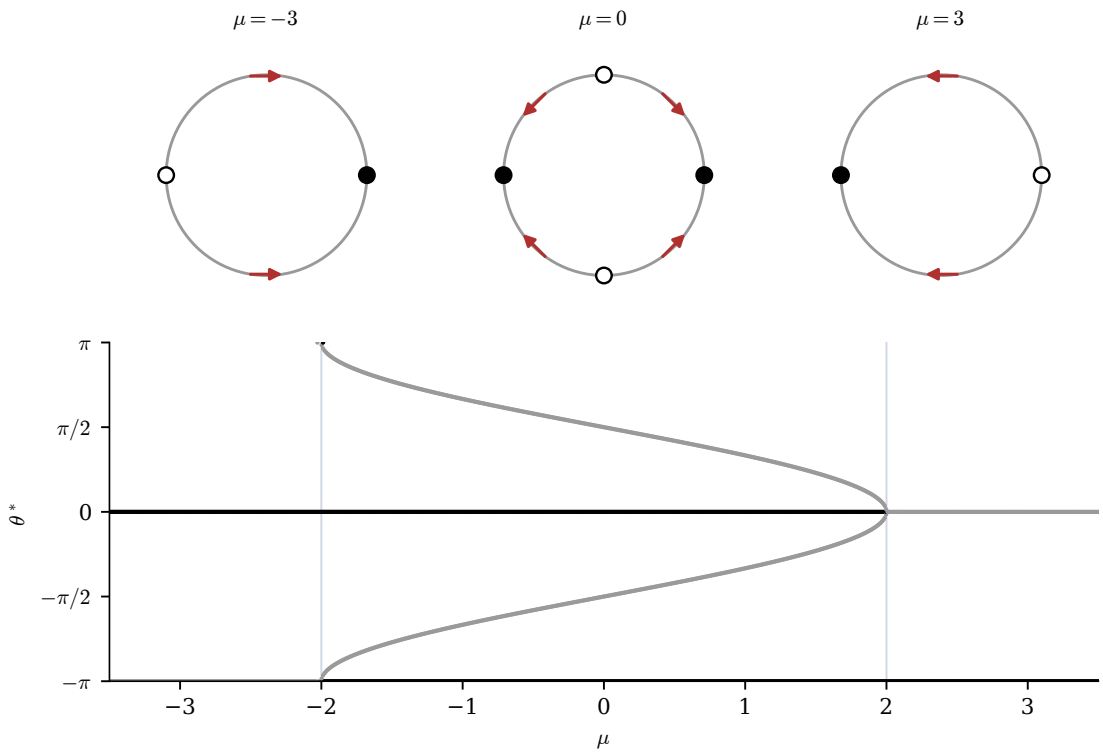


Figure 3: 4.3.3: portraits for $\mu = -3, 0, 3$ and bifurcation diagram (vertical lines at $\mu = \pm 2$).

Bifurcations: there are no saddle-nodes or pitchforks. At $\mu = 1$ the two singular points collide at $\theta = \pi$ with the stable fixed point (indeed $f'(\pi) = \frac{1}{1-\mu} \to -\infty$ as $\mu \to 1^+$ and at $\mu = 1$, $f = \tan\frac{\theta}{2}$ is singular at π); for $\mu < 1$ the point π is unstable and flanked by two attracting singularities. At $\mu = -1$ the same thing happens at $\theta = 0$ (where $f = -\cot\frac{\theta}{2}$). These are *non-smooth* transitions at $\mu = \pm 1$, where the vector field ceases to be well defined on the circle. ■

4.3.5 $\dot\theta = \mu + \cos\theta + \cos 2\theta$.

Solution. With $c = \cos\theta$ and $\cos 2\theta = 2c^2 - 1$: $f = 2c^2 + c + \mu - 1$. Fixed points: $2c^2 + c + (\mu - 1) = 0$, i.e.

$$c_\pm = \frac{-1 \pm s}{4}, \qquad s = \sqrt{9 - 8\mu},$$

real only for $\mu \le \frac{9}{8}$. Each admissible $c \in [-1, 1]$ gives two fixed points $\theta = \pm\cos^{-1} c$. Admissibility: $c_+ \le 1 \iff \mu \ge -2$; $c_- \ge -1 \iff \mu \ge 0$ (and $c_+ \ge -1$, $c_- \le 1$ always).

Stability: $f'(\theta) = -\sin\theta\,(1 + 4\cos\theta)$, and $1 + 4c_\pm = \pm s$. Hence $f'(+\cos^{-1} c_+) = -\sin\theta\, s < 0$ (stable) and $f'(-\cos^{-1} c_+) > 0$ (unstable); $f'(+\cos^{-1} c_-) = +\sin\theta\, s > 0$ (unstable) and $f'(-\cos^{-1} c_-) < 0$ (stable). (Because f is even, the fixed points $\pm\theta^*$ always have opposite stability.) So:

- $\mu > \frac{9}{8}$: no fixed points; $f > 0$ everywhere (its minimum is $\mu - \frac{9}{8}$ at $\cos\theta = -\frac{1}{4}$): steady counterclockwise rotation.
- $0 < \mu < \frac{9}{8}$: **four** fixed points: $\theta = +\cos^{-1} c_+$ (stable), $-\cos^{-1} c_+$ (unstable), $-\cos^{-1} c_-$ (stable), $+\cos^{-1} c_-$ (unstable). (For $\mu = 0.5$: stable at $+72°, -144°$, unstable at $-72°, +144°$.)

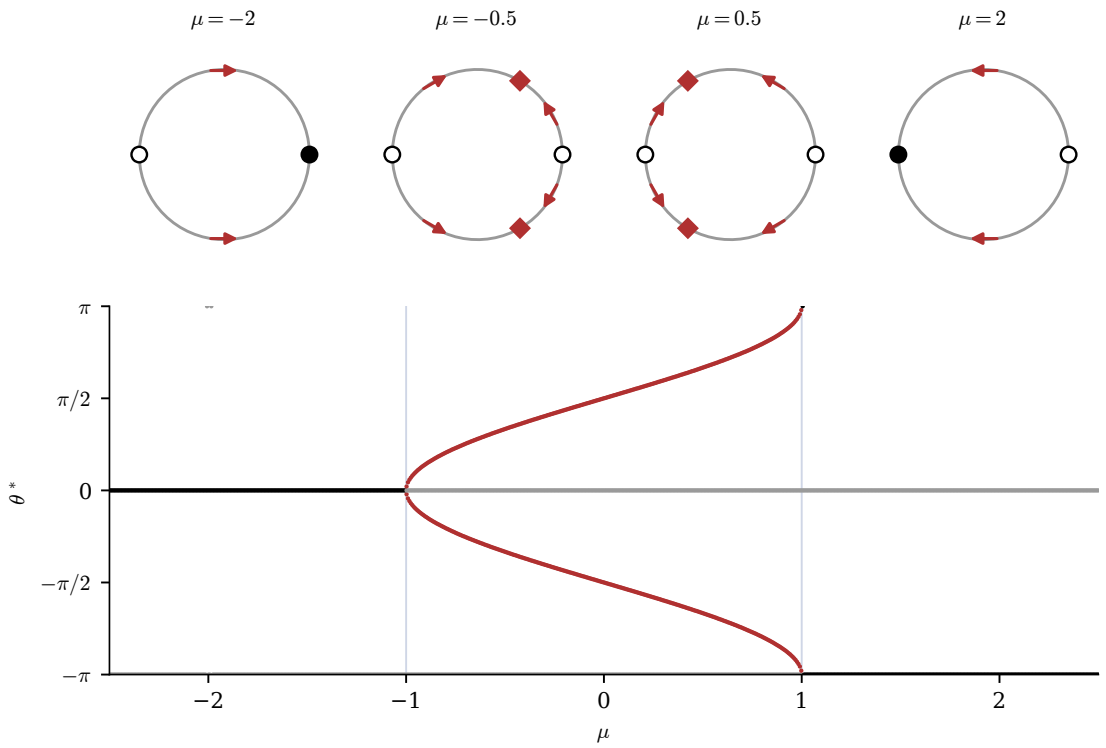


Figure 4: 4.3.4: portraits for $\mu = -2, -0.5, 0.5, 2$ (red diamonds: attracting singularities) and bifurcation diagram.

- $-2 < \mu < 0$: only $\pm\cos^{-1} c_+$ (stable at $+$, unstable at $-$).
- $\mu < -2$: no fixed points; $f < 0$, steady clockwise rotation.

Bifurcations (all saddle-node): at $\mu = \frac{9}{8}$, a stable/unstable pair is created at $\theta = \pm\cos^{-1}(-\frac{1}{4}) \approx \pm 104.5°$ (two simultaneous saddle-nodes, by the symmetry); at $\mu = 0$ the stable point $-\cos^{-1} c_-$ and the unstable point $+\cos^{-1} c_-$ collide at $\theta = \pi$ and annihilate; at $\mu = -2$ the pair $\pm\cos^{-1} c_+$ collide at $\theta = 0$ and annihilate. ■

4.3.6 $\dot\theta = \mu + \sin\theta + \cos 2\theta$.

Solution. With $s = \sin\theta$ and $\cos 2\theta = 1 - 2s^2$: $f = -2s^2 + s + \mu + 1$. Fixed points: $2s^2 - s - (\mu+1) = 0$, i.e.

$$s_\pm = \frac{1 \pm q}{4}, \qquad q = \sqrt{9 + 8\mu},$$

real for $\mu \geq -\frac{9}{8}$. Admissibility: $s_+ \leq 1 \iff \mu \leq 0$; $s_- \geq -1 \iff \mu \leq 2$. Each admissible s gives $\theta = \sin^{-1} s$ and $\pi - \sin^{-1} s$. Now $f'(\theta) = \cos\theta\,(1 - 4\sin\theta)$ with $1 - 4s_\pm = \mp q$. So:

- for s_+: $\theta = \sin^{-1} s_+$ ($\cos > 0$) has $f' < 0$: **stable**; $\theta = \pi - \sin^{-1} s_+$ is **unstable**;
- for s_-: $\theta = \sin^{-1} s_-$ is **unstable**; $\theta = \pi - \sin^{-1} s_-$ is **stable**.

Regimes: $\mu < -\frac{9}{8}$: none ($f < 0$ everywhere; clockwise rotation). $-\frac{9}{8} < \mu < 0$: four fixed points. $0 < \mu < 2$: only the two from s_- (one stable, one unstable). $\mu > 2$: none ($f > 0$; counterclockwise rotation).

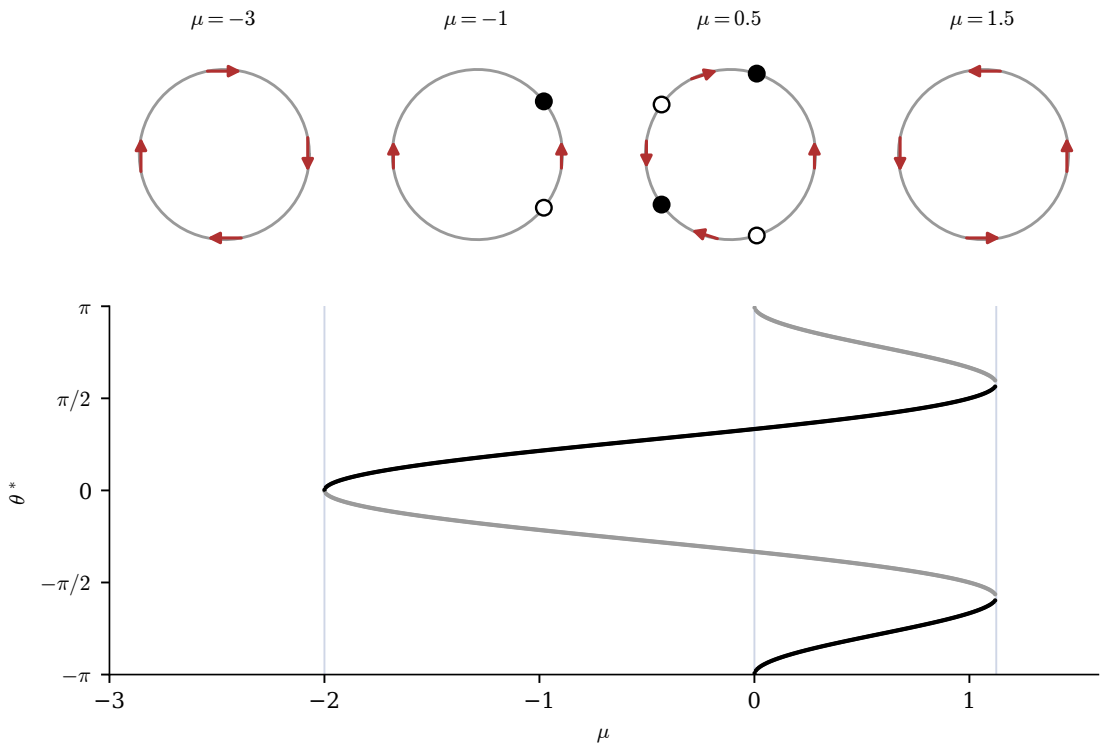


Figure 5: 4.3.5: portraits for $\mu = -3, -1, 0.5, 1.5$ and bifurcation diagram (lines at $\mu = -2, 0, \frac{9}{8}$).

Bifurcations (all saddle-node): $\mu = -\frac{9}{8}$: two simultaneous saddle-nodes at $s = \frac{1}{4}$, i.e. $\theta = \sin^{-1}\frac{1}{4} \approx 14.5°$ and $165.5°$; $\mu = 0$: the stable point $\sin^{-1} s_+$ and the unstable point $\pi - \sin^{-1} s_+$ collide at $\theta = \frac{\pi}{2}$ and annihilate; $\mu = 2$: the pair from s_- collides at $\theta = -\frac{\pi}{2}$ and annihilates. ■

4.3.7 $\dot{\theta} = \dfrac{\sin\theta}{\mu + \sin\theta}$.

Solution. Zeros: $\theta = 0, \pi$ (for $\mu \neq 0$). $f'(\theta) = \dfrac{\mu\cos\theta}{(\mu+\sin\theta)^2}$, so $f'(0) = \frac{1}{\mu}$ and $f'(\pi) = -\frac{1}{\mu}$.

- $\mu > 1$ (smooth): $\theta^* = 0$ **unstable**, $\theta^* = \pi$ **stable**. $\mu < -1$: 0 **stable**, π **unstable**.
- $0 < \mu < 1$: the denominator vanishes at $\sin\theta = -\mu$, i.e. $\theta_a = -\sin^{-1}\mu$ and $\theta_r = \pi + \sin^{-1}\mu$ (both in the lower half circle). Still 0 is **unstable** and π **stable**. Near θ_a (where $\cos > 0$) the flow runs into the singularity from both sides: *attracting*; near θ_r ($\cos < 0$) it runs away from it on both sides: *repelling*. Hence: on the upper half circle $f > 0$ and the flow runs from 0 to π; just beyond π the flow reverses ($f < 0$) and returns to π, up to the repelling singularity θ_r; beyond θ_r, $f > 0$ and the flow runs into θ_a; between θ_a and 0, $f < 0$, so the flow also runs back into θ_a.
- $-1 < \mu < 0$: the same portrait rotated by π: 0 **stable**, π **unstable**, singularities at $\sin\theta = |\mu|$ (upper half).
- $\mu = 0$: $f = \sin\theta/\sin\theta \equiv 1$ (removable singularities at $0, \pi$): uniform rotation, no fixed points.

Bifurcations (non-smooth): as $\mu \to 0^+$ the unstable fixed point 0 merges with the attracting

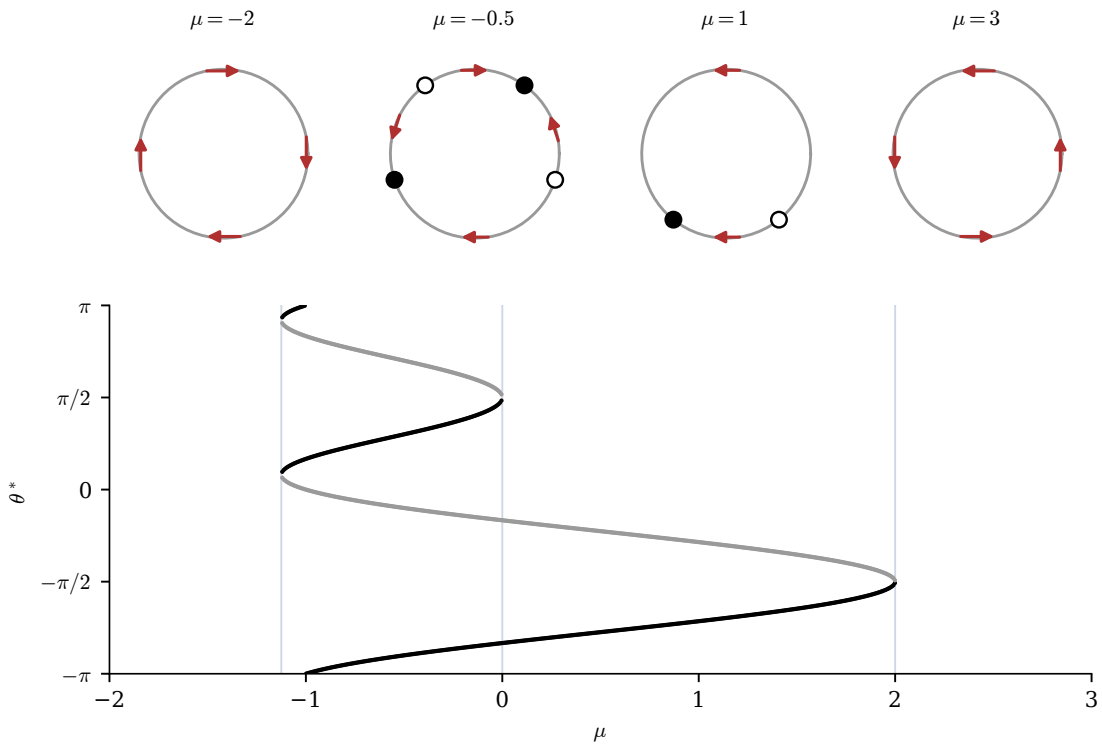


Figure 6: 4.3.6: portraits for $\mu = -2, -0.5, 1, 3$ and bifurcation diagram (lines at $\mu = -\frac{9}{8}, 0, 2$).

singularity at $-\sin^{-1}\mu$, and the stable point π merges with the repelling singularity at $\pi + \sin^{-1}\mu$; at $\mu = 0$ both pairs annihilate, leaving $f \equiv 1$. At $\mu = \pm 1$ the attracting and repelling singularities collide (at $\theta = \mp\frac{\pi}{2}$) and disappear for $|\mu| > 1$, leaving a smooth field. ■

4.3.8 $\dot\theta = \dfrac{\sin 2\theta}{1 + \mu \sin\theta}$.

Solution. The denominator vanishes only if $\sin\theta = -1/\mu$, which needs $|\mu| \geq 1$. Zeros: $\sin 2\theta = 0$, i.e. $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. At a zero, $f' = \dfrac{2\cos 2\theta}{1 + \mu\sin\theta}$, so

$$f'(0) = f'(\pi) = 2 > 0, \qquad f'(\tfrac{\pi}{2}) = -\frac{2}{1+\mu}, \qquad f'(\tfrac{3\pi}{2}) = -\frac{2}{1-\mu}.$$

- $|\mu| < 1$ (smooth): $0, \pi$ **unstable**; $\frac{\pi}{2}, \frac{3\pi}{2}$ **stable**: the same portrait as $\dot\theta = \sin 2\theta$.
- $\mu > 1$: $f'(\frac{3\pi}{2}) = \frac{2}{\mu - 1} > 0$, so $\frac{3\pi}{2}$ becomes **unstable**; the denominator now vanishes at $\theta_s = -\sin^{-1}(1/\mu)$ and $\pi + \sin^{-1}(1/\mu)$, one on each side of $\frac{3\pi}{2}$. These are **attracting singularities**: the flow leaves $\frac{3\pi}{2}$ and runs into them (and the flow from the outside also runs into them). The points $0, \pi$ remain unstable and $\frac{\pi}{2}$ stable.
- $\mu < -1$: the mirror image (rotation by π): $\frac{\pi}{2}$ becomes unstable, flanked by two attracting singularities.

Bifurcations (non-smooth "singular pitchforks"): at $\mu = 1$ the stable point $\frac{3\pi}{2}$ becomes unstable and emits two singular points (born at $\theta = \frac{3\pi}{2}$, where at $\mu = 1$ one has $f \approx -4/(\theta - \frac{3\pi}{2})$);

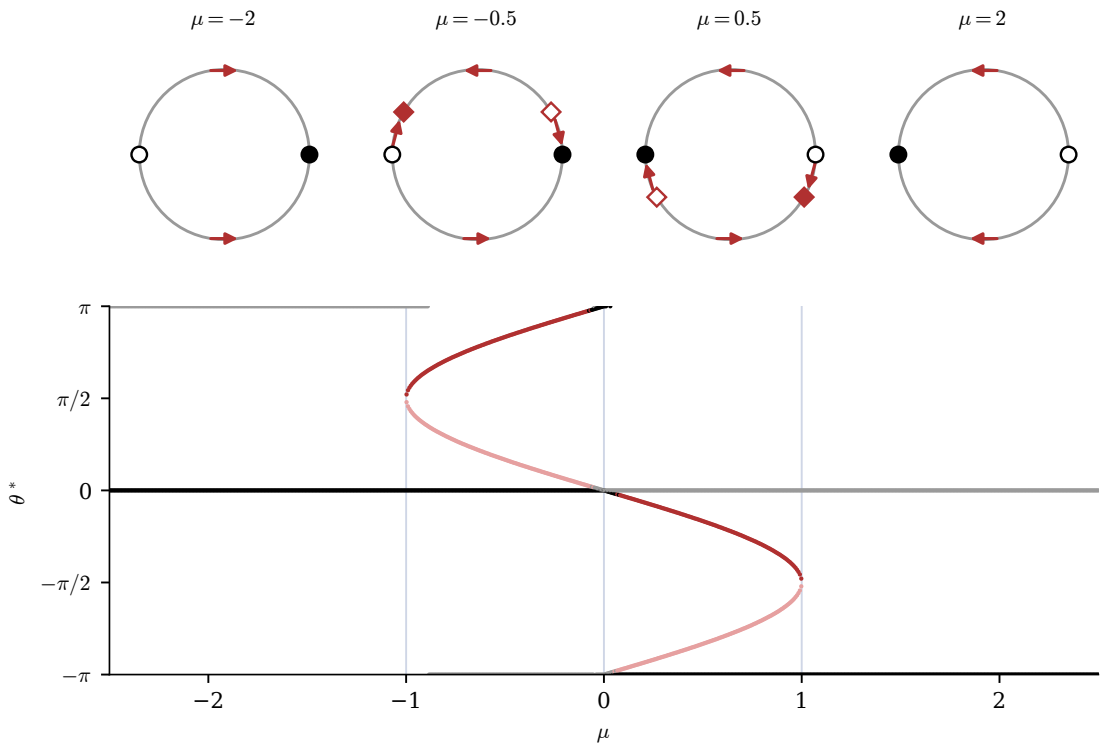


Figure 7: 4.3.7: portraits for $\mu = -2, -0.5, 0.5, 2$ (filled red diamond: attracting singularity; open: repelling) and bifurcation diagram (lines at $\mu = -1, 0, 1$).

similarly at $\mu = -1$ at $\theta = \frac{\pi}{2}$. They resemble a subcritical pitchfork, but the new branches are poles of the vector field rather than fixed points. ■

4.3.9 *(Alternative derivation of the scaling law) For $\dot{x} = r + x^2$ near a saddle-node bottleneck, suppose $x = r^a u$, $t = r^b \tau$ with $u, \tau = O(1)$. (a) Show the equation becomes $r^{a-b}\,\mathrm{d}u/\mathrm{d}\tau = r + r^{2a}u^2$. (b) Assume all terms are of the same order in r and derive $a = \frac{1}{2}$, $b = -\frac{1}{2}$.*

Solution. **(a)** $\dot{x} = \dfrac{\mathrm{d}x}{\mathrm{d}t} = \dfrac{r^a\,\mathrm{d}u}{r^b\,\mathrm{d}\tau} = r^{a-b}\dfrac{\mathrm{d}u}{\mathrm{d}\tau}$, and $r + x^2 = r + r^{2a}u^2$.

(b) All three terms are comparable when their powers of r agree: $a - b = 1 = 2a$. Hence $a = \frac{1}{2}$ and $b = a - 1 = -\frac{1}{2}$. The equation becomes $\mathrm{d}u/\mathrm{d}\tau = 1 + u^2$, free of r. So the characteristic length is $x \sim r^{1/2}$ and the characteristic time is $t \sim r^{-1/2}$: the bottleneck time scales as $T \sim O(r^{-1/2})$, in agreement with $T = \pi/\sqrt{r}$ (4.3.1). ■

4.3.10 *(Nongeneric scaling laws) For $\dot{x} = r + x^{2n}$ ($n > 1$ an integer) show $T_{\text{bottleneck}} \approx c\,r^b$ and determine b and c.*

Solution. Scale $x = r^a u$, $t = r^b \tau$: $r^{a-b}u' = r + r^{2na}u^{2n}$. Balancing, $a - b = 1 = 2na$, so

$$a = \frac{1}{2n}, \qquad b = a - 1 = -\frac{2n-1}{2n} = \boxed{-1 + \frac{1}{2n}}.$$

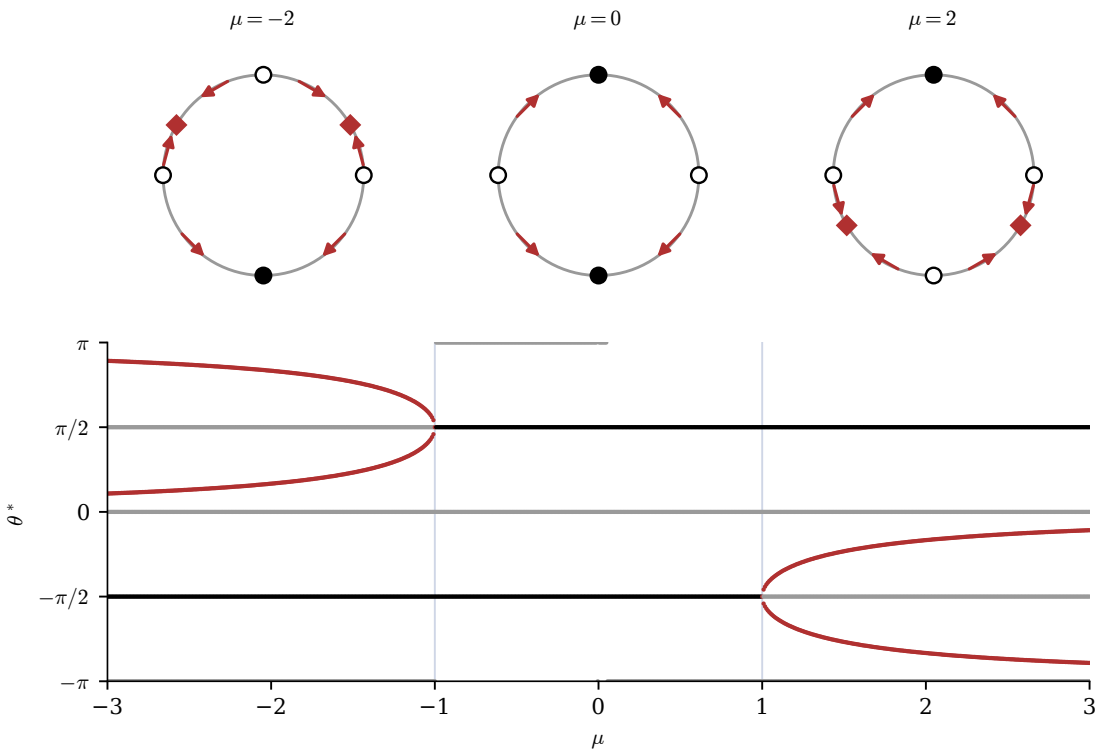


Figure 8: 4.3.8: portraits for $\mu=-2,0,2$ and bifurcation diagram (lines at $\mu=\pm1$).

Directly, $T=\int_{-\infty}^{\infty}\frac{\mathrm{d}x}{r+x^{2n}}$; with $x=r^{1/2n}u$, $r+x^{2n}=r(1+u^{2n})$:

$$T=r^{\frac{1}{2n}-1}\,c,\qquad c=\int_{-\infty}^{\infty}\frac{\mathrm{d}u}{1+u^{2n}}.$$

Evaluating c. Let $I_0=\int_0^{\infty}\frac{\mathrm{d}x}{1+x^{2n}}$ and integrate $1/(1+z^{2n})$ around the boundary of the pie-slice $0\le\arg z\le\pi/n$, $|z|\le R$. Along the ray $z=\rho\,\mathrm{e}^{i\pi/n}$, $z^{2n}=\rho^{2n}$, so that side contributes $-\mathrm{e}^{i\pi/n}I_0$ as $R\to\infty$; the arc contribution vanishes ($\sim R^{1-2n}$). The only pole inside is $z_0=\mathrm{e}^{i\pi/2n}$, with residue $\frac{1}{2nz_0^{2n-1}}=-\frac{z_0}{2n}$ (since $z_0^{2n}=-1$). Thus

$$(1-\mathrm{e}^{i\pi/n})I_0=2\pi i\Big(-\frac{\mathrm{e}^{i\pi/2n}}{2n}\Big).$$

Since $1-\mathrm{e}^{i\pi/n}=-2i\,\mathrm{e}^{i\pi/2n}\sin\frac{\pi}{2n}$, we get $I_0=\dfrac{\pi}{2n\sin\frac{\pi}{2n}}$, and by evenness

$$c=2I_0=\boxed{\frac{\pi/n}{\sin\left(\frac{\pi}{2n}\right)}}.$$

Check: $n=1$ gives $c=\pi$, $b=-\frac{1}{2}$ (4.3.1); for $n=5$ (the equation of 2.5.2) $c=\frac{\pi/5}{\sin(\pi/10)}\approx 2.03$. For larger n the exponent $b\to-1$: the bottleneck gets slower, $T\sim r^{-1+1/2n}$. ■

4.4 Overdamped Pendulum

4.4.1 *(Validity of the overdamped limit) Find the conditions under which* $mL^2\ddot{\theta}+b\dot{\theta}+mgL\sin\theta=\Gamma$ *may be approximated by* $b\dot{\theta}+mgL\sin\theta=\Gamma$.

Solution. Nondimensionalize with $\tau=\dfrac{mgL}{b}\,t$ (the time scale set by the overdamped balance between the damping and gravity torques) and $\gamma=\dfrac{\Gamma}{mgL}$. Then $\dot{\theta}=\frac{mgL}{b}\theta'$, $\ddot{\theta}=\left(\frac{mgL}{b}\right)^2\theta''$, and dividing the equation by mgL:

$$\varepsilon\,\theta''+\theta'+\sin\theta=\gamma,\qquad \varepsilon=\frac{m^2gL^3}{b^2}.$$

The inertia term is negligible after a short initial transient (Section 3.5) when

$$\boxed{\varepsilon=\frac{m^2gL^3}{b^2}\ll 1\quad\Longleftrightarrow\quad b\gg mL\sqrt{gL}}$$

(for any γ, since the other terms are $O(1)$). Physically, the inertial relaxation time mL^2/b must be much shorter than the time b/mgL over which the overdamped motion evolves; i.e. viscous damping must dominate inertia (large damping, small mass or length). The transient lasts $\Delta\tau\sim\varepsilon$. ■

4.4.2 *(Understanding* $\sin\theta(t)$*) Sketch* $\sin\theta(t)$ *for typical solutions of* $\theta'=\gamma-\sin\theta$ *for different* γ*, including* $\gamma\approx 1$ *and* $\gamma\gg 1$*. What physical quantity is proportional to* $\sin\theta$ *for the pendulum?*

Solution. For $\gamma<1$ the solution tends to the stable fixed point $\theta^*=\sin^{-1}\gamma$, so $\sin\theta\to\gamma$ (constant). For $\gamma>1$ the pendulum rotates forever with period $T=2\pi/\sqrt{\gamma^2-1}$ (Section 4.3). The solution is explicit: with $u=\tan\frac{\theta}{2}$ the equation becomes $u'=\frac{\gamma}{2}\left[(u-\frac{1}{\gamma})^2+\frac{\gamma^2-1}{\gamma^2}\right]$, so

$$\tan\frac{\theta}{2}=\frac{1+\omega\tan\left(\frac{\omega\tau}{2}+c\right)}{\gamma},\quad \omega=\sqrt{\gamma^2-1},\qquad \sin\theta=\frac{2u}{1+u^2}.$$

Waveforms (Fig. 9, left column; time in units of the period):

- $\gamma\gtrsim 1$: the pendulum lingers for most of the cycle near $\theta=\pi/2$ (the ghost of the saddle-node, where $\theta'=\gamma-\sin\theta$ is tiny) and then swings rapidly through the rest of the circle. So $\sin\theta(t)$ sits near $+1$ for most of the period and has a *narrow dip* down to -1 once per period: a very non-sinusoidal, spiky waveform.
- Increasing γ: the plateau shortens and the dip widens; the waveform becomes more and more sinusoidal.
- $\gamma\gg 1$: $\theta'\approx\gamma$, so $\theta\approx\gamma\tau$ and $\sin\theta(t)\approx\sin(\gamma\tau)$, a nearly pure sinusoid of amplitude 1 and period $2\pi/\gamma$ (with a small distortion $O(1/\gamma)$).

Physical meaning: $mgL\sin\theta$ is the **gravitational torque** on the pendulum; so $\sin\theta(t)$ is proportional to the torque exerted by gravity (equivalently, in the balance $b\dot{\theta}=\Gamma-mgL\sin\theta$, it is the torque not used for driving the rotation; for a Josephson junction it is the supercurrent $I_c\sin\phi$).■

4.4.3 *(Understanding* $\dot{\theta}(t)$*) Redo 4.4.2 for* $\dot{\theta}(t)$.

Solution. In the overdamped limit $\theta'=\gamma-\sin\theta$, so the graph of $\theta'(t)$ is the graph of $\sin\theta(t)$ flipped upside down and shifted up by γ (Fig. 9, right column).

- $\gamma \gtrsim 1$: $\theta' \approx \gamma - 1 \approx 0$ for most of the period (slow creep), with a sharp *pulse* up to $\gamma + 1 \approx 2$ once per period while the pendulum whips around: a periodic pulse train.
- $\gamma \gg 1$: $\theta' \approx \gamma - \sin(\gamma\tau)$: a large constant rotation rate with a small sinusoidal ripple of amplitude 1.

The mean angular velocity over a period is $\langle\theta'\rangle = 2\pi/T = \sqrt{\gamma^2 - 1}$ (smaller than γ, because of the slow stretch). Physically $\dot\theta$ is the angular velocity, which for the pendulum is $\propto$ the viscous torque $b\dot\theta$; for the Josephson junction it is proportional to the voltage $V = \frac{\hbar}{2e}\dot\phi$ (4.6.1). ■

4.4.4 *(Torsional spring) The spring adds a torque $-k\theta$: $b\dot\theta + mgL\sin\theta = \Gamma - k\theta$. (a) Is this a well-defined vector field on the circle? (b) Nondimensionalize. (c) Long-run behavior? (d) Show that many bifurcations occur as k goes from 0 to ∞; of what type?*

Solution. **(a)** $\dot\theta = \big(\Gamma - k\theta - mgL\sin\theta\big)/b$. Replacing θ by $\theta + 2\pi$ changes the right side by $-2\pi k/b \neq 0$ (if $k \neq 0$). So the right side is *not* 2π-periodic: this is **not** a vector field on the circle. θ must be treated as a real number (the spring winds up as the pendulum rotates, so θ and $\theta + 2\pi$ are physically different states): the system is a flow on the *line*.

(b) With $\tau = \frac{mgL}{b}t$, $\gamma = \frac{\Gamma}{mgL}$, $\kappa = \frac{k}{mgL}$:

$$\theta' = \gamma - \kappa\theta - \sin\theta.$$

(c) For $\kappa > 0$, $f(\theta) = \gamma - \kappa\theta - \sin\theta \to \mp\infty$ as $\theta \to \pm\infty$, so there is at least one fixed point, and a flow on the line cannot oscillate: every solution tends monotonically to a stable fixed point. The pendulum **eventually stops** (it does not rotate forever, unlike $\kappa = 0$, $\gamma > 1$): it settles at an equilibrium where the spring torque plus gravity balance the applied torque.

(d) Fixed points are the intersections of the line $y = \gamma - \kappa\theta$ with the curve $y = \sin\theta$, and $f'(\theta) = -\kappa - \cos\theta$. For $\kappa > 1$, $f' < 0$ everywhere: f is strictly decreasing, so there is exactly one fixed point, stable. For $\kappa < 1$ the line crosses the sine curve in the window $\theta \in [(\gamma-1)/\kappa, (\gamma+1)/\kappa]$ about $2/(\pi\kappa)$ times, and as κ increases the line steepens and pairs of intersections (one stable, one unstable) are destroyed when the line becomes tangent to $y = \sin\theta$, i.e. when $\cos\theta = -\kappa$ and $\sin\theta = \gamma - \kappa\theta$. So as κ goes from 0 to ∞ there are **many saddle-node bifurcations**: infinitely many accumulate as $\kappa \to 0^+$ (Fig. 10, $\gamma = 2$), until a single stable fixed point remains for $\kappa > 1$. (For $\kappa = 0$ and $\gamma > 1$ there are none and the pendulum rotates; for small κ it rotates until the spring has wound far enough to reach the first fixed point.) ■

4.5 Fireflies

Model: $\dot\Theta = \Omega$ (stimulus), $\dot\theta = \omega + Af(\Theta - \theta)$ (firefly). With $\phi = \Theta - \theta$:

$$\dot\phi = \Omega - \omega - Af(\phi) = \Delta - Af(\phi), \qquad \Delta := \Omega - \omega.$$

4.5.1 *(Triangle wave) $f(\phi) = \phi$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\pi - \phi$ on $[\frac{\pi}{2}, \frac{3\pi}{2}]$, extended 2π-periodically. (a) Graph f. (b) Range of entrainment. (c) Phase difference ϕ^* when locked. (d) T_{drift}.*

Solution. **(a)** A triangle wave of period 2π with maximum $\frac{\pi}{2}$ at $\phi = \frac{\pi}{2}$ and minimum $-\frac{\pi}{2}$ at $\phi = -\frac{\pi}{2}$ (and $\frac{3\pi}{2}$), zero at $\phi = 0, \pi$ (Fig. 11, left).

(b) Entrainment means a fixed point $\dot\phi = 0$ exists: $f(\phi^*) = \Delta/A$. Since f ranges over $[-\frac{\pi}{2}, \frac{\pi}{2}]$,

$$\boxed{|\Omega - \omega| \le \frac{\pi A}{2}}.$$

(c) For $|\nu| \le \frac{\pi}{2}$ with $\nu = \Delta/A$: $f(\phi) = \nu$ has the solutions $\phi = \nu$ (on the rising branch) and $\phi = \pi - \nu$ (falling branch). Since $\dot\phi = \Delta - Af$ has derivative $-Af'(\phi)$, the rising-branch point ($f' = +1$) is **stable** and the falling-branch point ($f' = -1$) is **unstable**. The phase-locked difference is

$$\boxed{\phi^* = \frac{\Omega - \omega}{A}} \pmod{2\pi},$$

(the unstable one is $\pi - \Delta/A$). When $\Omega > \omega$ the firefly lags the stimulus ($\phi^* > 0$) and when $\Omega < \omega$ it leads. The two fixed points merge in a saddle-node at $\phi = \pm\frac{\pi}{2}$ when $|\Delta| = \frac{\pi A}{2}$.

(d) For $|\Delta| > \frac{\pi A}{2}$ (say $\Delta > 0$), $\dot\phi > 0$ and the drift period is $T_{\text{drift}} = \int_0^{2\pi} \frac{\mathrm{d}\phi}{\Delta - Af(\phi)}$, integrating over one period $[-\frac{\pi}{2}, \frac{3\pi}{2}]$:

$$\int_{-\pi/2}^{\pi/2} \frac{\mathrm{d}\phi}{\Delta - A\phi} = \frac{1}{A}\ln\frac{\Delta + \pi A/2}{\Delta - \pi A/2}, \qquad \int_{\pi/2}^{3\pi/2} \frac{\mathrm{d}\phi}{\Delta - A(\pi - \phi)} = \frac{1}{A}\ln\frac{\Delta + \pi A/2}{\Delta - \pi A/2}$$

(the second with $\psi = \phi - \pi$). Hence

$$\boxed{T_{\text{drift}} = \frac{2}{A}\ln\frac{|\Delta| + \pi A/2}{|\Delta| - \pi A/2}}.$$

Checks: for $|\Delta| \gg A$, $T \approx \frac{2}{A}\cdot\frac{\pi A}{|\Delta|} = \frac{2\pi}{|\Delta|}$ (uniform drift). As $|\Delta| \to \frac{\pi A}{2}{}^+$, T diverges only *logarithmically*, in contrast to the sinusoidal case, where $T = 2\pi/\sqrt{\Delta^2 - A^2}$ diverges like an inverse square root (Fig. 11, right). The reason: at its maximum the triangle wave has a corner, $f \approx \frac{\pi}{2} - |\phi - \frac{\pi}{2}|$, so the bottleneck is $\dot\phi \approx \varepsilon + A|\psi|$ rather than $\varepsilon + A\psi^2/2$. ■

4.5.2 *(General response function) Redo 4.5.1 assuming only that f is smooth, 2π-periodic, with a single maximum $f_{\max}$ (at ϕ_M) and a single minimum $f_{\min}$ on $[-\pi, \pi]$.*

Solution. Entrainment range. A fixed point of $\dot\phi = \Delta - Af(\phi)$ exists iff Δ/A lies in the range of f:

$$\boxed{Af_{\min} \le \Omega - \omega \le Af_{\max}}$$

(not necessarily symmetric about $\Omega = \omega$ if $f_{\max} \ne -f_{\min}$).

Locked phase. For Δ/A strictly inside the range, $f(\phi) = \Delta/A$ has exactly two solutions per period (by monotonicity of f on the two arcs between ϕ_m and ϕ_M): ϕ_s^* on the arc where f is increasing ($f' > 0$, giving $\frac{\mathrm{d}\dot\phi}{\mathrm{d}\phi} = -Af' < 0$, **stable**) and ϕ_u^* on the arc where f is decreasing (**unstable**). Explicitly $\phi_s^* = (f|_{\text{inc}})^{-1}(\Delta/A)$. Since f has no closed form we cannot say more; for $f = \sin$ one gets $\phi^* = \sin^{-1}(\Delta/A)$.

Edge of the range. At $\Delta = Af_{\max}$ the two fixed points merge at ϕ_M, and at $\Delta = Af_{\min}$ at ϕ_m: a saddle-node bifurcation, after which phase locking is lost (drift).

Drift period. Outside the range, $\dot\phi$ has a fixed sign and

$$T_{\text{drift}} = \int_{-\pi}^{\pi} \frac{\mathrm{d}\phi}{\Delta - Af(\phi)}.$$

Near the edge, $\Delta = Af_{\max} + \varepsilon$ with $\varepsilon > 0$ small, expand around ϕ_M: $\Delta - Af(\phi) \approx \varepsilon + \frac{1}{2}A|f''(\phi_M)|\,\psi^2$, so

$$T_{\text{drift}} \approx \int_{-\infty}^{\infty} \frac{\mathrm{d}\psi}{\varepsilon + \frac{1}{2}A|f''(\phi_M)|\psi^2} = \pi\sqrt{\frac{2}{A|f''(\phi_M)|\,\varepsilon}} \;\propto\; \left(\Delta - Af_{\max}\right)^{-1/2},$$

the generic square-root law of a saddle-node bottleneck (the sine function being the special case $f''(\frac{\pi}{2}) = -1$, $\varepsilon = \Delta - A$, which gives $T \approx \pi\sqrt{2/(A\varepsilon)}$, matching $2\pi/\sqrt{\Delta^2 - A^2}$ for $\Delta \approx A$). For $|\Delta| \gg A$, $T \approx 2\pi/|\Delta|$. ■

4.5.3 *(Excitable systems) $\dot{\theta} = \mu + \sin\theta$ with μ slightly less than 1. (a) Show that it has the two properties of an excitable system. What plays the role of "rest state" and "threshold"? (b) Sketch $V(t) = \cos\theta(t)$ for various initial conditions.*

Solution. **(a)** For $\mu < 1$ there are two fixed points, $\sin\theta^* = -\mu$:

$$\theta_b = \pi + \sin^{-1}\mu \;\; (f' = \cos\theta_b < 0,\ \text{stable}), \qquad \theta_a = 2\pi - \sin^{-1}\mu \;\; (f' > 0,\ \text{unstable}),$$

located just on either side of $\theta = \frac{3\pi}{2}$, at $\frac{3\pi}{2} \mp \sqrt{2(1-\mu)}$ for $\mu \approx 1$ (close together). Between them $f = \mu + \sin\theta < 0$; elsewhere $f > 0$.

(a) *Unique, globally attracting rest state.* Every initial condition except the unstable point θ_a itself converges to the stable fixed point θ_b (on the circle, the only possible long-time behaviors are fixed points or rotation, and with a stable fixed point present and $\mu < 1$ there is no rotation). So θ_b is the **rest state**.

(b) *A large enough stimulus gives a long excursion.* A perturbation that moves θ from θ_b to a value between θ_b and θ_a (below threshold) simply relaxes straight back to θ_b. But if the perturbation carries θ beyond the unstable point θ_a, then $f > 0$ and the phase runs all the way around the circle (passing $\theta = 2\pi$) before returning to θ_b from the other side. The unstable fixed point θ_a is the **threshold**. The excursion is the same for any supra-threshold stimulus ("all-or-none"), and since μ is close to 1 the passage near the rest/threshold region is slow, so the spike is followed by a long recovery.

(b) $V = \cos\theta$ at rest equals $\cos\theta_b \approx 0^-$. Fig. 12 (with $\mu = 0.95$) shows: for sub-threshold initial conditions ($\theta_b < \theta_0 < \theta_a$) V makes only a small excursion and relaxes back; for supra-threshold initial conditions V rises to $+1$ (at $\theta = 2\pi$), falls through 0 to -1 (at $\theta = 3\pi$) and returns to rest: an identical spike of height 1 and undershoot to -1, regardless of how far above threshold the stimulus pushed θ_0 (only the initial delay differs). This is the caricature of a neuron's action potential. ■

4.6 Superconducting Josephson Junctions

In the overdamped limit the phase obeys (Eq. 4.6.7) $\phi' = \gamma - \sin\phi$ with $\gamma = I/I_c$ and dimensionless time $\tau = \frac{2eI_cR}{\hbar}t$; the voltage is $V = \frac{\hbar}{2e}\dot{\phi}$.

4.6.1 *(Current and voltage oscillations) Take $\beta = 0$. (a) Sketch $I_c\sin\phi(t)$ for I/I_c slightly greater than 1 and for $I/I_c \gg 1$. (b) Sketch $V(t)$.*

Solution. **(a)** This is the same equation as in 4.4.2 with $\gamma = I/I_c$ (Fig. 9, left; Fig. 13). For $I/I_c \gtrsim 1$ the phase sits near $\phi = \frac{\pi}{2}$ for most of the cycle, so the supercurrent $I_c\sin\phi$ is nearly constant at $+I_c$, with brief, sharp dips to $-I_c$ once per period: $T = \frac{2\pi}{\sqrt{\gamma^2 - 1}}$ in τ, i.e. a very slow cycle.

For $I/I_c \gg 1$, $I_c \sin\phi(t) \approx I_c \sin(\frac{2eIR}{\hbar}t)$: a small, nearly sinusoidal ac supercurrent of amplitude I_c at frequency $\frac{2eIR}{\hbar}$.

(b) $V = \frac{\hbar}{2e}\dot{\phi} = I_c R\,\phi' = I_c R\,(\gamma - \sin\phi) = R(I - I_c \sin\phi)$ (Ohm's law for the resistive channel, since the total current is $I = I_c \sin\phi + V/R$). For $I/I_c \gtrsim 1$: $V \approx 0$ for most of the cycle and a sharp *voltage pulse* of height up to $\approx 2I_cR$ once per period. For $I/I_c \gg 1$: $V \approx IR$ with a small sinusoidal ripple of amplitude I_cR. The dc (time-averaged) voltage is

$$\langle V\rangle = \frac{\hbar}{2e}\cdot\frac{2\pi}{T_{\text{real}}} = I_cR\sqrt{\gamma^2-1} = R\sqrt{I^2-I_c^2} \qquad (I > I_c),$$

the I–V characteristic of the overdamped junction. ■

4.6.2 *(Computer work) Check the sketches of 4.6.1 by integrating (4.6.7) numerically and plotting $I_c \sin\phi(t)$ and $V(t)$.*

Solution. I integrated $\phi' = \gamma - \sin\phi$, $\phi(0) = 0$, with an adaptive Runge–Kutta method (tolerances 10^{-10}) for $\gamma = 1.1$ and $\gamma = 3$ (Fig. 13); the dashed line in the voltage panels marks the predicted mean $\langle V\rangle/I_cR = \sqrt{\gamma^2-1}$. The results agree with 4.6.1: for $\gamma = 1.1$ the supercurrent is nearly flat at I_c with narrow dips and the voltage consists of narrow pulses (mean $0.46\,I_cR$); for $\gamma = 3$ both are nearly sinusoidal about 0 and $\gamma\,I_cR = IR$ respectively. The measured period (time for ϕ to advance by 2π) agrees with $2\pi/\sqrt{\gamma^2-1}$ to at least six digits in each case. ■

4.6.3 *(Washboard potential) (a) Find the potential for (4.6.7) and show it is not single-valued on the circle. (b) Graph it as a function of the real number ϕ for various I/I_c. (c) Effect of increasing I?*

Solution. **(a)** $\phi' = \gamma - \sin\phi = -\dfrac{\mathrm{d}V}{\mathrm{d}\phi}$ gives

$$V(\phi) = -\gamma\phi - \cos\phi.$$

Then $V(\phi + 2\pi) = V(\phi) - 2\pi\gamma \neq V(\phi)$ unless $\gamma = 0$: not single-valued on the circle (consistent with 4.1.8: $\oint(\gamma - \sin\phi)\,\mathrm{d}\phi = 2\pi\gamma \neq 0$).

(b) V is a cosine "washboard" tilted down to the right with slope $-\gamma$ (Fig. 14). $V' = -\gamma + \sin\phi$, $V'' = \cos\phi$. For $\gamma < 1$, V has local minima at $\phi = \sin^{-1}\gamma + 2\pi n$ (stable fixed points) and local maxima at $\pi - \sin^{-1}\gamma + 2\pi n$ (unstable), the barrier height being $\Delta V = 2\sqrt{1-\gamma^2} - 2\gamma\cos^{-1}\gamma$. For $\gamma = 1$ the minima and maxima merge into inflection points ($V' \le 0$ with $V' = 0$ at isolated points). For $\gamma > 1$, $V' < 0$ everywhere: no extrema.

(c) Increasing I tilts the washboard more steeply: the wells become shallower (the barrier $\Delta V \to 0$ as $I \to I_c^-$), the phase particle gets pushed over, and at $I = I_c$ the wells disappear in a saddle-node bifurcation. For $I > I_c$ the particle slides down the washboard forever (the phase runs, giving a nonzero mean voltage $\propto \langle\dot{\phi}\rangle$), slowing down as it crosses each corrugation, which is the origin of the nonuniform rotation seen in 4.6.1. ■

4.6.4 *(Resistively loaded array) Two identical overdamped junctions (each in parallel with a resistor r) in series, in parallel with a load R, driven by a dc bias I_b. (a) Relate I_b, I_a, I_R. (b) Show $I_a = I_c\sin\phi_k + V_k/r$. (c) Express V_k. (d) Show $I_b = I_c\sin\phi_k + \frac{\hbar}{2er}\dot{\phi}_k + \frac{\hbar}{2eR}(\dot{\phi}_1 + \dot{\phi}_2)$. (e) Derive $\dot{\phi}_k = \Omega + a\sin\phi_k + K\sum_{j=1}^{2}\sin\phi_j$ and find Ω, a, K.*

Solution. **(a)** Kirchhoff's current law at the node where the bias current splits between the array and the load: $\boxed{I_b = I_a + I_R}$.

(b) Junction k ($k = 1, 2$) is an overdamped junction in parallel with a resistor r (Fig. 4.6.3 with $C = 0$ and $R \to r$): the current through it is the sum of the supercurrent and the current through the resistor, $I_a = I_c \sin\phi_k + V_k/r$ (the same series current I_a flows through both junctions).

(c) By the Josephson voltage–phase relation, $V_k = \dfrac{\hbar}{2e}\dot{\phi}_k$.

(d) The load is in parallel with the array, so the load voltage equals the total array voltage: $I_R R = V_1 + V_2$, i.e. $I_R = \dfrac{\hbar}{2eR}(\dot{\phi}_1 + \dot{\phi}_2)$. Then (a), (b), (c) give, for $k = 1, 2$,

$$I_b = I_c \sin\phi_k + \frac{\hbar}{2er}\dot{\phi}_k + \frac{\hbar}{2eR}(\dot{\phi}_1 + \dot{\phi}_2).$$

(e) Add the two equations ($k = 1, 2$) and let $S = \dot{\phi}_1 + \dot{\phi}_2$:

$$2I_b = I_c(\sin\phi_1 + \sin\phi_2) + \frac{\hbar}{2e}\Big(\frac{1}{r} + \frac{2}{R}\Big)S \implies \frac{\hbar}{2e}S = \frac{rR}{R+2r}\Big[2I_b - I_c\sum_{j=1}^{2}\sin\phi_j\Big].$$

Substitute in the kth equation, using $\frac{\hbar}{2eR}S = \frac{r}{R+2r}\big[2I_b - I_c\sum\sin\phi_j\big]$:

$$\frac{\hbar}{2er}\dot{\phi}_k = I_b\Big(1 - \frac{2r}{R+2r}\Big) - I_c\sin\phi_k + \frac{rI_c}{R+2r}\sum_{j=1}^{2}\sin\phi_j.$$

Multiplying by $2er/\hbar$:

$$\boxed{\dot{\phi}_k = \Omega + a\sin\phi_k + K\sum_{j=1}^{2}\sin\phi_j}$$

with

$$\Omega = \frac{2er}{\hbar}\frac{R\,I_b}{R+2r}, \qquad a = -\frac{2erI_c}{\hbar}, \qquad K = \frac{2e\,r^2 I_c}{\hbar\,(R+2r)}.$$

(Here $a < 0$ in this orientation; it is the "self" term $-I_c\sin\phi_k$ of an isolated junction, while $K > 0$ is the coupling through the load.) Checks: $R \to \infty$ gives $K \to 0$, $\Omega \to \frac{2er}{\hbar}I_b$ and $\dot{\phi}_k = \frac{2er}{\hbar}(I_b - I_c\sin\phi_k)$, the uncoupled overdamped junction; $R \to 0$ (short circuit) gives $\Omega = 0$ and, adding the two equations, $\dot{\phi}_1 + \dot{\phi}_2 = 0$, i.e. no voltage across the array, as it must be. ■

4.6.5 *(N junctions, resistive load) Show that the governing equations can be written* $\dfrac{\mathrm{d}\phi_k}{\mathrm{d}\tau} = \Omega + a\sin\phi_k + \dfrac{1}{N}\sum_{j=1}^{N}\sin\phi_j$, $k = 1, \ldots, N$, *and give* Ω, a *and* τ.

Solution. Now $I_b = I_a + I_R$, $I_a = I_c\sin\phi_k + \frac{\hbar}{2er}\dot{\phi}_k$ for each k, and the load voltage is the sum of the N junction voltages, $I_R = \frac{\hbar}{2eR}\sum_j \dot{\phi}_j$. Summing the N equations $I_b = I_c\sin\phi_k + \frac{\hbar}{2er}\dot{\phi}_k + \frac{\hbar}{2eR}\sum_j\dot{\phi}_j$:

$$NI_b = I_c\sum_j\sin\phi_j + \frac{\hbar}{2e}\Big(\frac{1}{r} + \frac{N}{R}\Big)\sum_j\dot{\phi}_j \implies \frac{\hbar}{2eR}\sum_j\dot{\phi}_j = \frac{r}{R+Nr}\Big[NI_b - I_c\sum_j\sin\phi_j\Big].$$

Hence

$$\dot\phi_k = \frac{2er}{\hbar}\Big[\frac{R\,I_b}{R+Nr} - I_c\sin\phi_k + \frac{rI_c}{R+Nr}\sum_{j=1}^{N}\sin\phi_j\Big].$$

Choose the dimensionless time so that the coupling coefficient is $1/N$:

$$\boxed{\tau = \frac{2eI_c\,r^2N}{\hbar\,(R+Nr)}\,t} \qquad (\text{so that } \tfrac{\mathrm{d}}{\mathrm{d}\tau} = \tfrac{\hbar(R+Nr)}{2eI_cr^2N}\tfrac{\mathrm{d}}{\mathrm{d}t}).$$

Then

$$\frac{\mathrm{d}\phi_k}{\mathrm{d}\tau} = \Omega + a\sin\phi_k + \frac{1}{N}\sum_j \sin\phi_j, \qquad \boxed{\Omega = \frac{R\,I_b}{N\,r\,I_c}}, \quad \boxed{a = -\Big(1+\frac{R}{Nr}\Big) = -\frac{R+Nr}{Nr}}.$$

(For $N=2$ these agree with 4.6.4 after the time rescaling: Ω/ω, a/ω, K/ω with $\omega = \frac{2eI_cr^2N}{\hbar(R+Nr)}$.) With this orientation of the Josephson relations a comes out negative; it measures the strength of the "self" term relative to the global coupling, and $|a| = 1 + \frac{R}{Nr}$. ■

4.6.6 *(N junctions, RLC load) The load is a resistor R in series with a capacitor C and inductor L. Write differential equations for ϕ_k and the load capacitor charge Q.*

Solution. The load branch carries the current $I_R = \dot Q$ (series elements). The array is in parallel with the load, so the array voltage equals the voltage drop across the load branch:

$$\frac{\hbar}{2e}\sum_{j=1}^{N}\dot\phi_j = L\ddot Q + R\dot Q + \frac{Q}{C}.$$

Kirchhoff's current law gives $I_b = I_a + \dot Q$ with $I_a = I_c\sin\phi_k + \frac{\hbar}{2er}\dot\phi_k$, so

$$\boxed{\frac{\hbar}{2er}\dot\phi_k = I_b - \dot Q - I_c\sin\phi_k, \qquad k = 1,\dots,N}.$$

To eliminate $\sum\dot\phi_j$ from the load equation, sum the previous equations: $\frac{\hbar}{2e}\sum_j\dot\phi_j = r\big[N(I_b-\dot Q) - I_c\sum_j\sin\phi_j\big]$. Then

$$\boxed{L\ddot Q + (R+Nr)\dot Q + \frac{Q}{C} = NrI_b - rI_c\sum_{j=1}^{N}\sin\phi_j}.$$

Together these are $N+2$ first-order equations for $(\phi_1,\dots,\phi_N,Q,\dot Q)$. Check: for $L=0$ and $C\to\infty$ (a pure resistive load, $\dot Q = I_R$) the second equation gives $\dot Q = \frac{r}{R+Nr}\big[NI_b - I_c\sum\sin\phi_j\big]$, and substituting into the first reproduces the equation of 4.6.5. ■

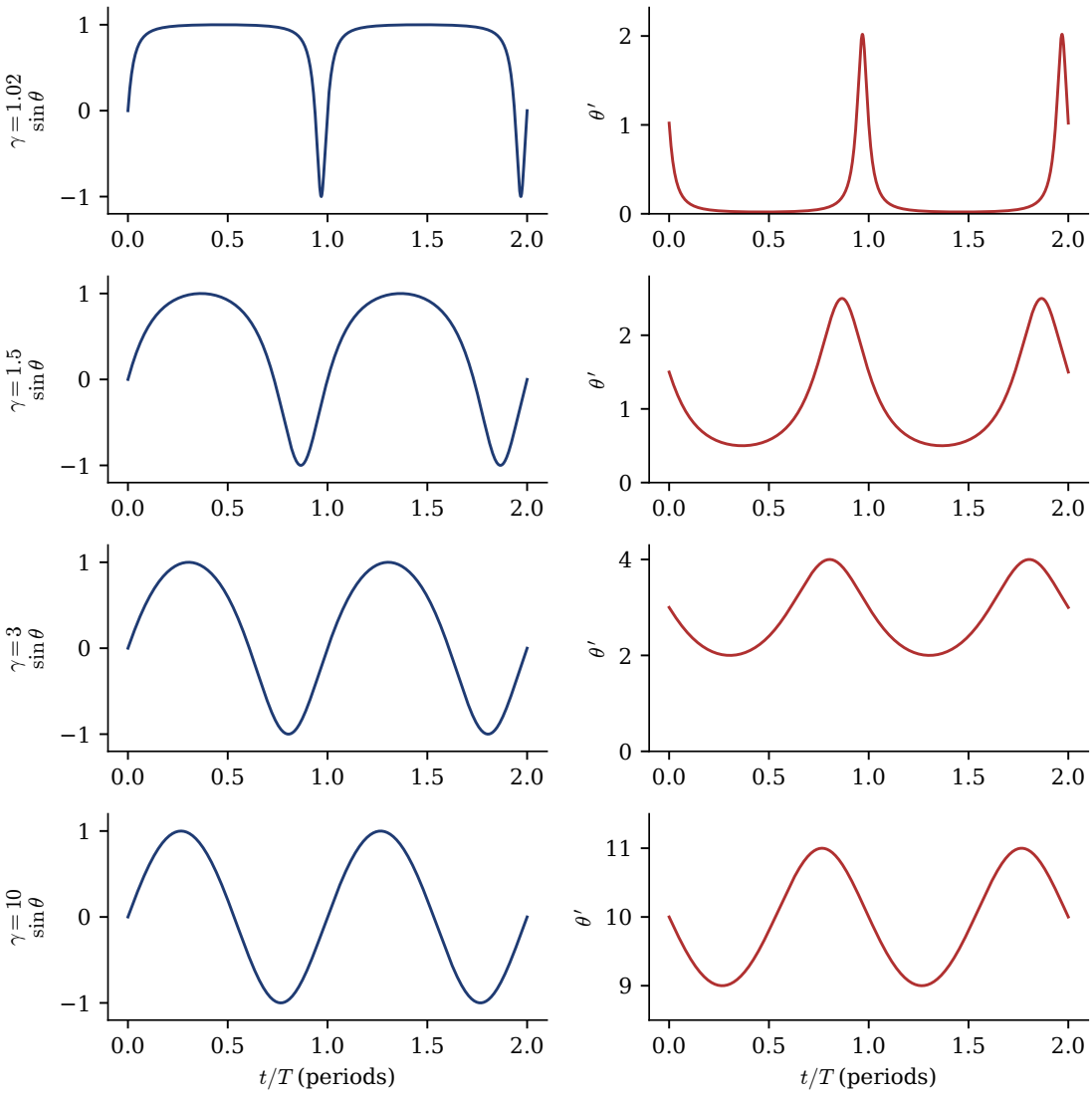


Figure 9: Solutions of $\theta' = \gamma - \sin\theta$ for $\gamma = 1.02, 1.5, 3, 10$, over two periods: $\sin\theta(t)$ (left) and $\theta'(t) = \gamma - \sin\theta$ (right).

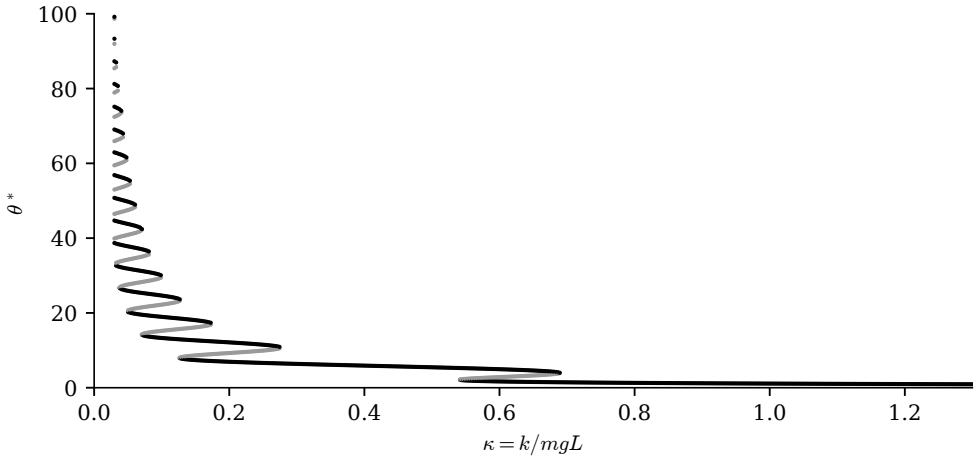


Figure 10: 4.4.4: fixed points of $\theta' = \gamma - \kappa\theta - \sin\theta$ versus κ for $\gamma = 2$ (black: stable, grey: unstable). Pairs are created/annihilated in saddle-node bifurcations.

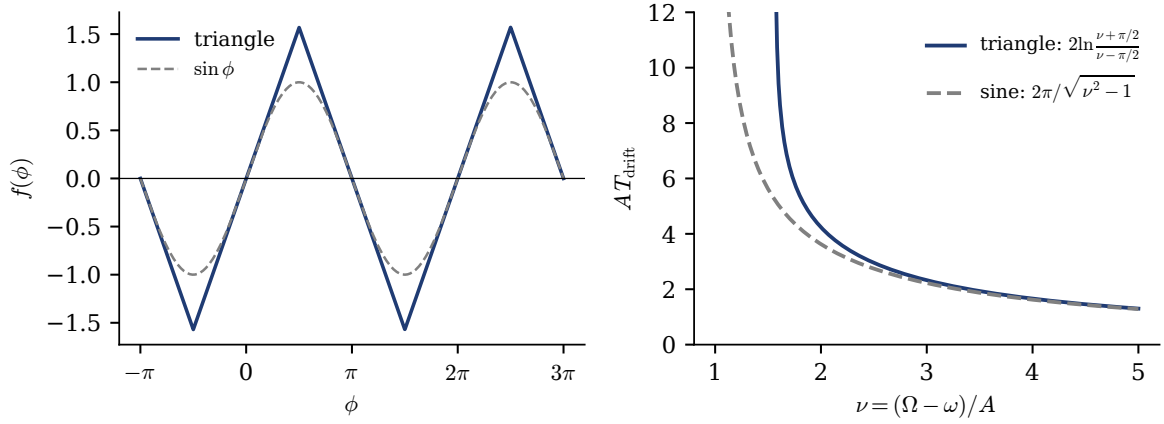


Figure 11: 4.5.1: the triangle wave f (left) and the drift period vs. detuning (right), compared with the sinusoidal model.

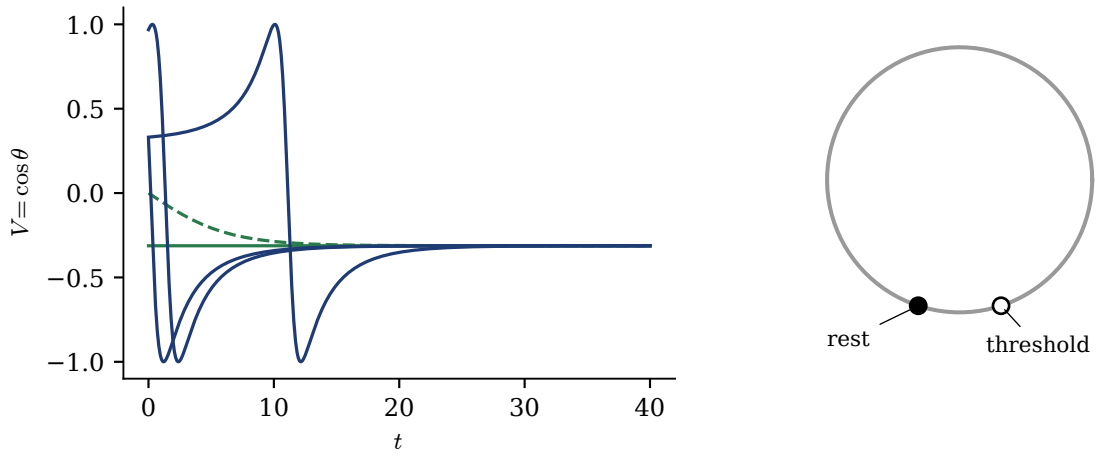


Figure 12: 4.5.3: $V(t) = \cos\theta(t)$ for $\mu = 0.95$ (green: rest and a sub-threshold start; blue: supra-threshold starts) and the rest/threshold points on the circle.

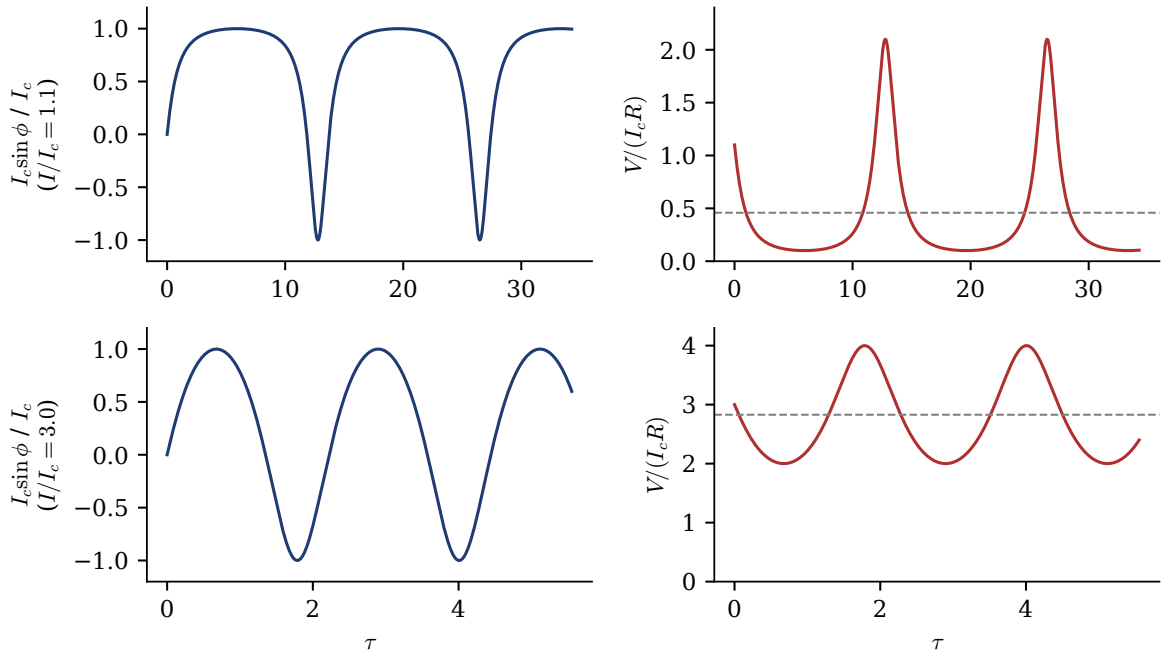


Figure 13: 4.6.2: numerical $I_c \sin\phi$ (units of I_c, left) and V (units of I_cR, right), for $I/I_c = 1.1$ (top) and 3 (bottom); dashed: predicted mean voltage.

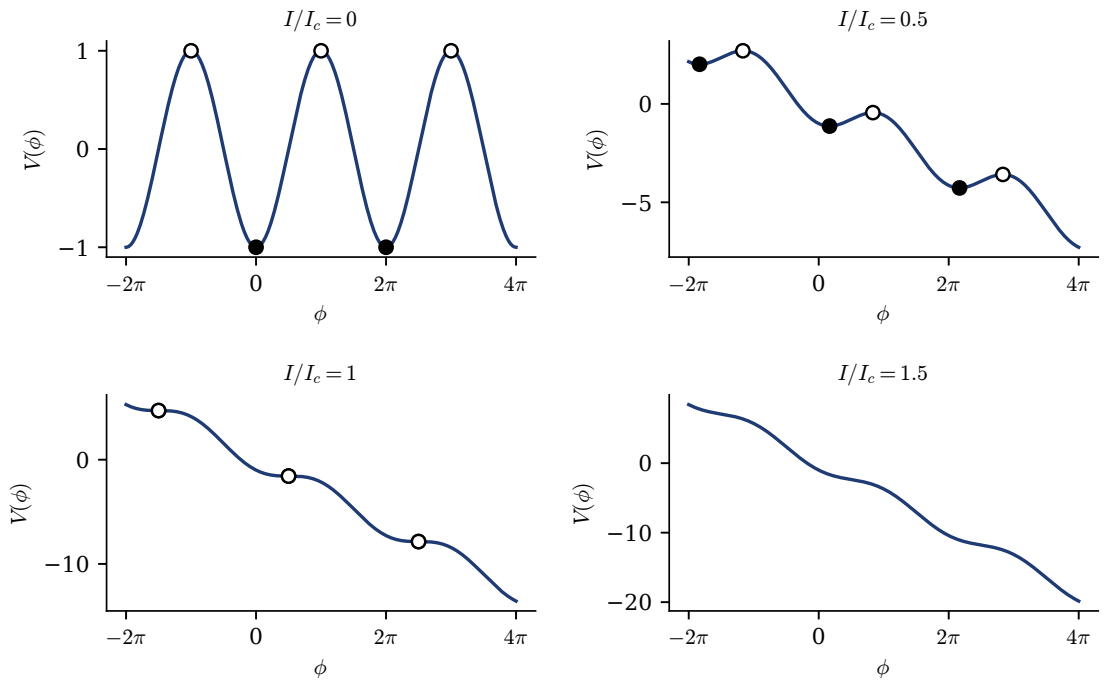


Figure 14: 4.6.3: washboard potential $V(\phi) = -\gamma\phi - \cos\phi$ for $I/I_c = 0, 0.5, 1, 1.5$ (filled: stable, open: unstable equilibria).